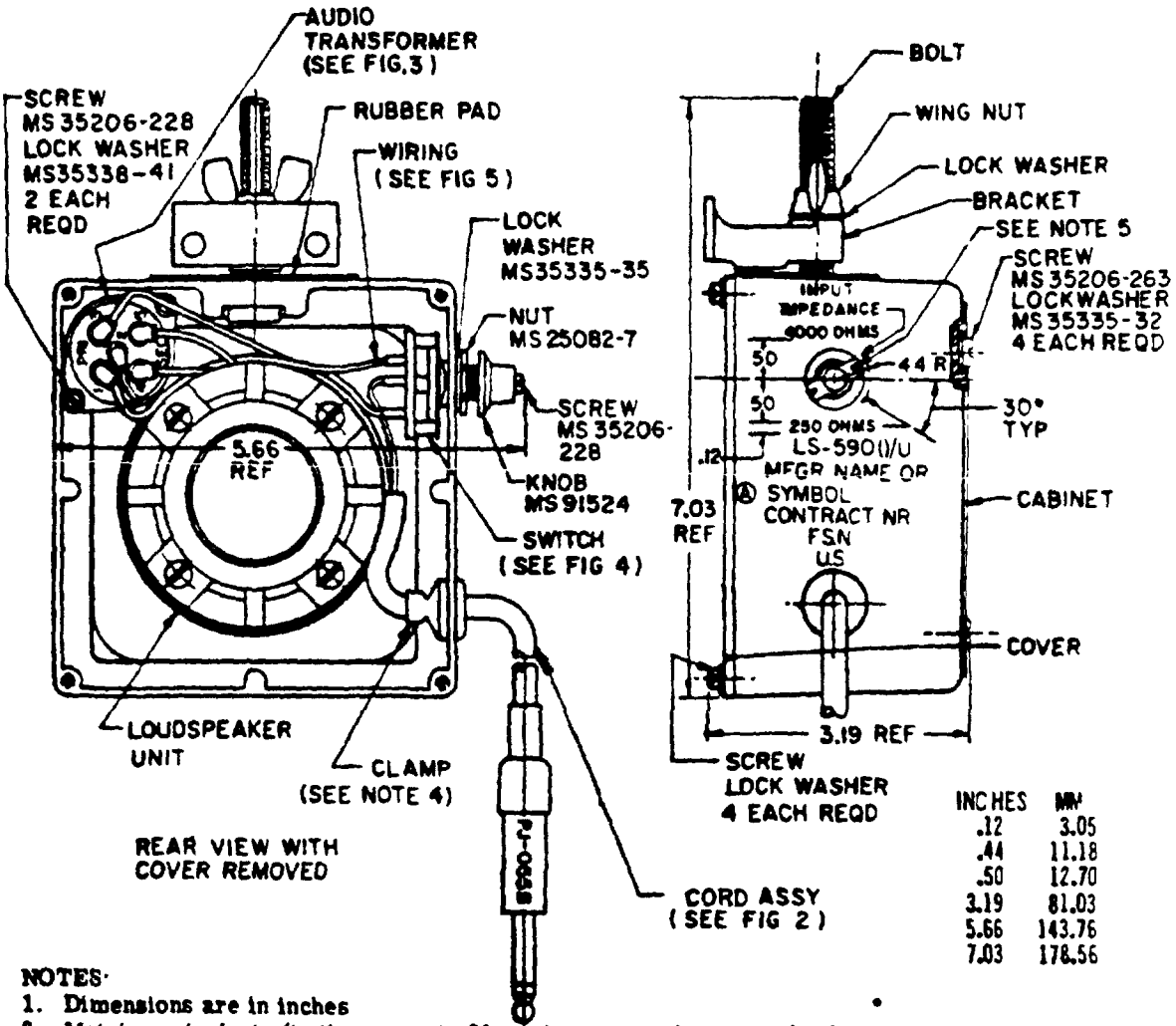
MILITARY SPECIFICATION SHEET

LOUDSPEAKER, PERMANENT MAGNET (ENCASED, 2-WATT, FUNGUS-, GUNBLAST-, AND IMMERSION-RESISTANT), TYPE LS-590( )/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the loudspeaker described herein shall consist of this document and the latest issue of Specification MIL-L-12632



INCHES	MM
.12	3.05
.44	11.18
.50	12.70
3.19	81.03
5.66	143.76
7.03	178.56

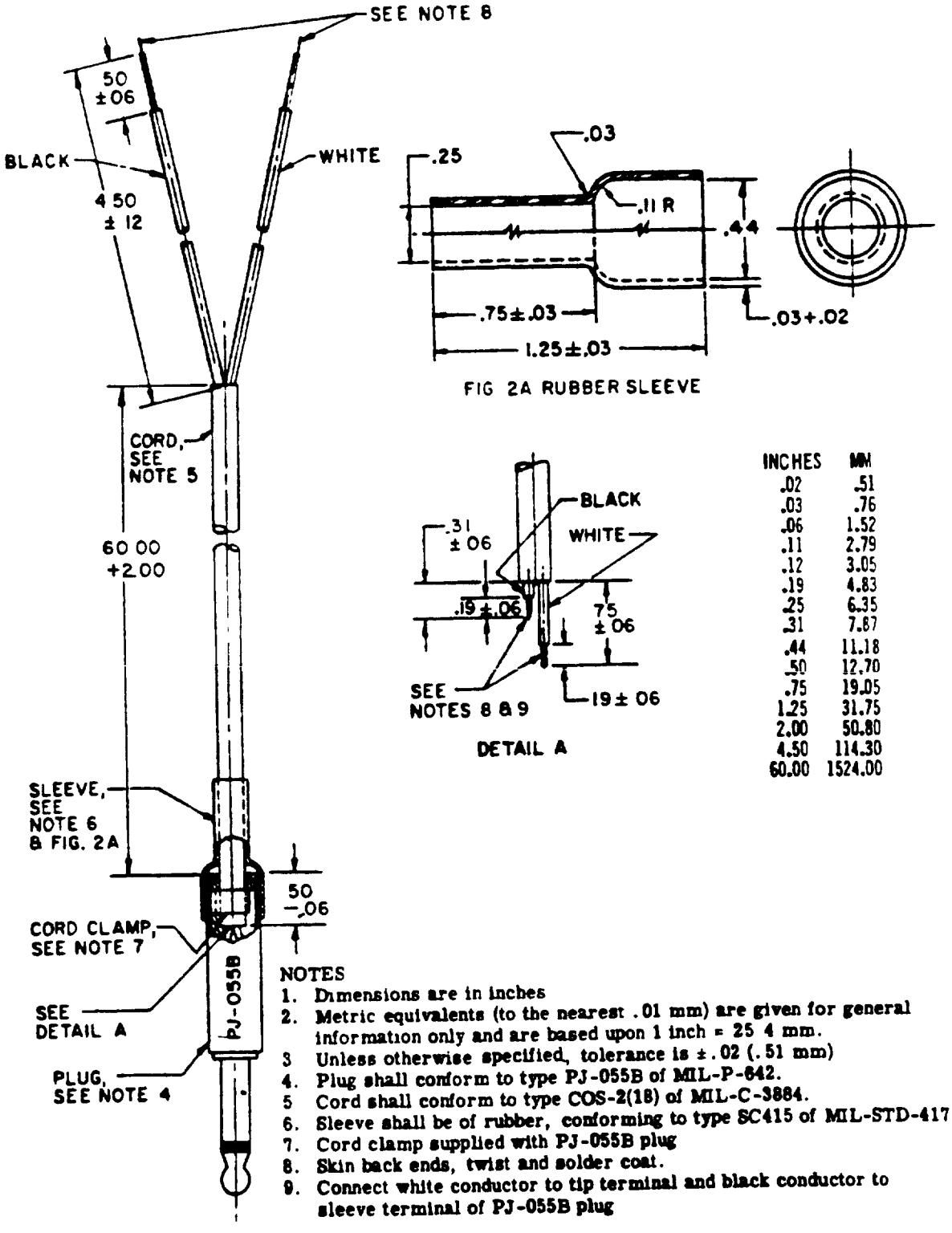
NOTES:
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.02 (.51 mm) and ±1/2° on angles
4. Steel clamp firmly clamped on cable jacket. Design optional. A strain relief bushing (e g Heyco) is a suitable substitute for the grommet/clamp combination
5. Drill .391 (9.93 mm) diameter hole for rotary switch

FIGURE 1. Loudspeaker LS-590()/U

(A) denotes changes.

FSC 5965

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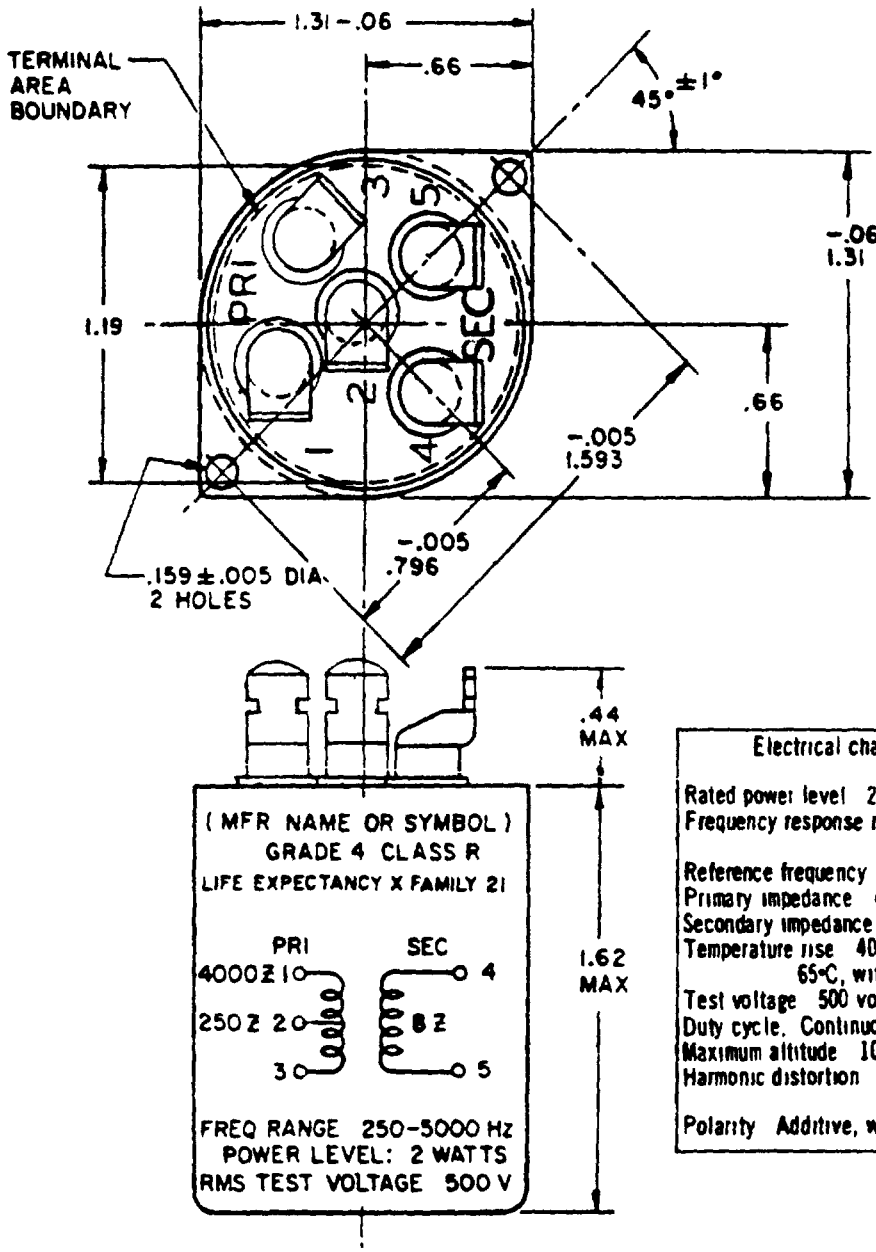


INCHES	MM
.02	.51
.03	.76
.06	1.52
.11	2.79
.12	3.05
.19	4.83
.25	6.35
.31	7.87
.44	11.18
.50	12.70
.75	19.05
1.25	31.75
2.00	50.80
4.50	114.30
60.00	1524.00

NOTES
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm.
3. Unless otherwise specified, tolerance is ±.02 (.51 mm)
4. Plug shall conform to type PJ-055B of MIL-P-642.
5. Cord shall conform to type COS-2(18) of MIL-C-3884.
6. Sleeve shall be of rubber, conforming to type SC415 of MIL-STD-417
7. Cord clamp supplied with PJ-055B plug
8. Skin back ends, twist and solder coat.
9. Connect white conductor to tip terminal and black conductor to sleeve terminal of PJ-055B plug

FIGURE 2. Cord assembly.

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INCHES	MM
.005	.13
.06	1.52
.159	4.04
.44	11.18
.66	16.76
.769	20.22
1.19	30.23
1.31	33.27
1.593	40.46
1.62	41.15

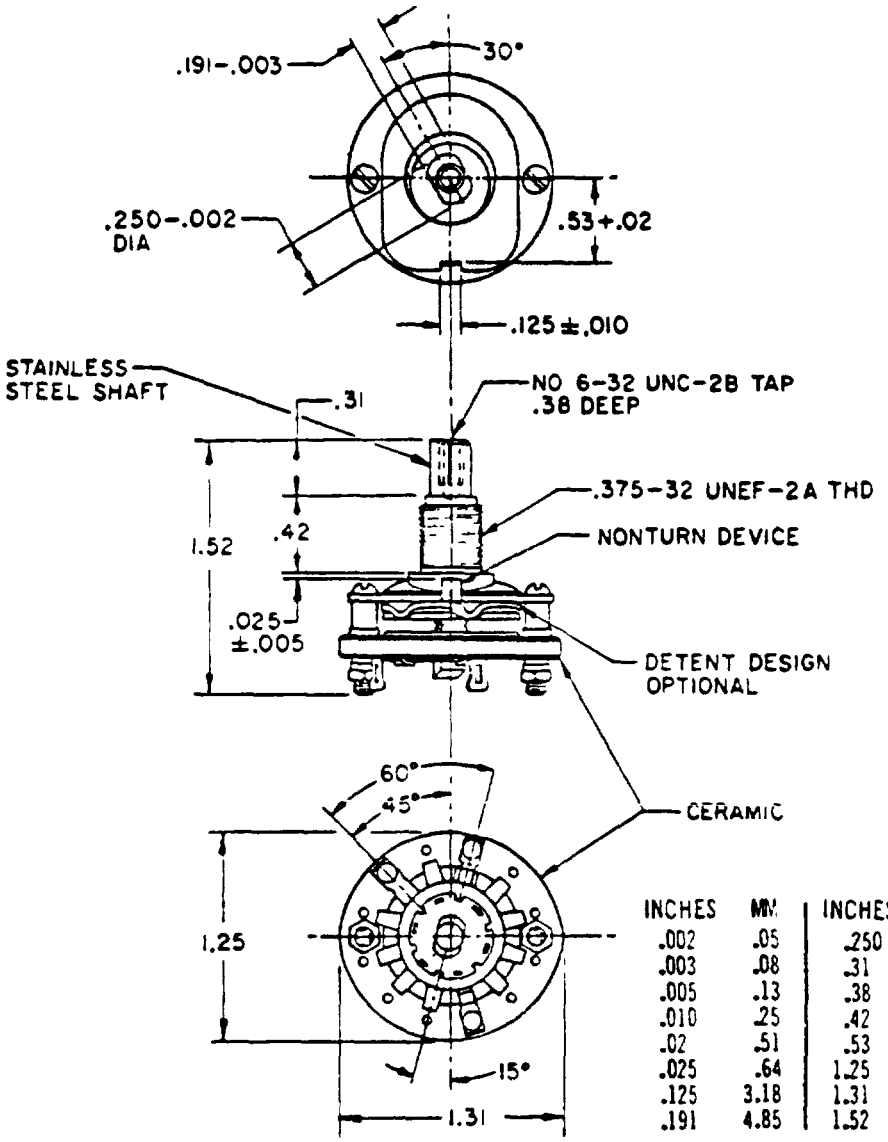
Electrical characteristics

Rated power level 2 watts
Frequency response range 250 to 5000 Hz at rated power
Reference frequency 1000 Hz
Primary impedance 4000 ±400Ω tapped at 250 ±25Ω.
Secondary impedance 8Ω
Temperature rise 40°C, max, above an ambient of 65°C, with rated power at 300 Hz.
Test voltage 500 volts rms.
Duty cycle, Continuous
Maximum altitude 10,000 feet.
Harmonic distortion Not more than 1 percent at rated power.
Polarity Additive, with terminals 3 and 4 connected.

NOTES
1 Dimensions are in inches
2 Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
3. Unless otherwise specified, tolerance is ±.02 (.51 mm).
4 Solder lug terminals, No. 18 AWG.
5 Markings shall be as shown

FIGURE 3. Audio transformer.

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INCHES	MM	INCHES	MM
.002	.05	.250	6.35
.003	.08	.31	7.87
.005	.13	.38	9.65
.010	.25	.42	10.67
.02	.51	.53	13.46
.025	.64	1.25	31.75
.125	3.18	1.31	33.27
.191	4.85	1.52	38.61

NOTES:
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
3. Unless otherwise specified, tolerances are ±.02 (.51 mm) and ±1/2°.
4. Construction-single section, single pole
5. Contacts shall be nonshorting
6. Stops shall be provided to limit rotation to two positions
7. Terminals-angle of bend 60° to 90°
8. Torque (rotational) At 25° +10° -5°C: 1.5 pound-inches min: 6 pound inches max
   At -55°C ±2°C 1.5 pound-inches min: 8 pound inches max
9. Life, rotational 10,000 rotational cycles at +125°C

FIGURE 4. Rotary switch.

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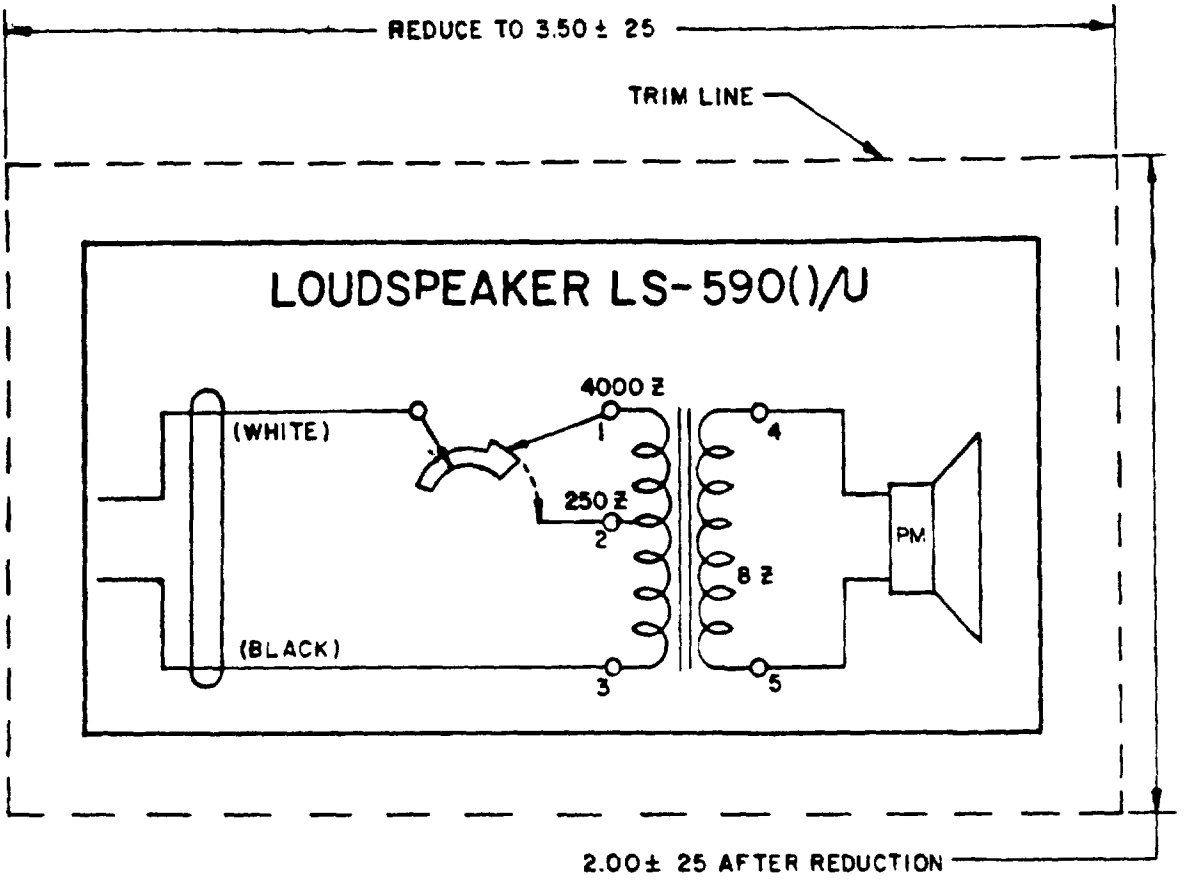


FIGURE 5. Schematic diagram and label size

REQUIREMENTS.

Design and construction (see figure 1):
Loudspeaker unit - Type LS-445/U of MIL-L-12606.
Cabinet - In accordance with MIL-L-12632.
Cord assembly - See figure 2.
Cord - Type COS-2(18) of MIL-C-3884.
Plug - Type PJ-055B of MIL-P-642.
Audio transformer - See figure 3.
Rotary switch - See figure 4.
Wiring - See figure 5.

Schematic-diagram label: In accordance with MIL-L-12632 and figure 5.

Frequency response test voltage (rms)
22.5 volts (250-ohm connection).
90.0 volts (4000-ohm connection).

Marking: See figure 1.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 80, 13
DSA - ES

User activities:
Army - ME
Navy - MC, CG, OS
Air Force -

Preparing activity
Army - EL

Agent:
DSA - ES

(Project 5965-0141-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-L-12632/1A LOUDSPEAKER,PERMANENT MAGNET (ENCASED,2-WATT, FUNGUS-, GUNELAST-,AND IMMERSION-RESISTANT),TYPE LS-590()/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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